

This note introduces the foundations of learning theory, including PAC learning, Rademacher complexity, and VC dimension. We begin with the finite, realizable PAC bound and then move to agnostic empirical-risk minimization. For infinite hypothesis classes, we replace raw cardinality with more informative capacity measures: Rademacher complexity, growth functions, VC dimension, and Sauer's lemma. Proofs are included so that each inequality can be traced to concentration, a union bound, symmetrization, or a counting argument.

1 Learning setup and the PAC question

1.1 Data, concepts, and hypotheses

Let $(\mathcal{X}, \mathcal{A})$ be an instance space and let $\mathcal{Y} = \{0, 1\}$ (or, equivalently, $\{-1, +1\}$). A target concept is a function $c : \mathcal{X} \rightarrow \mathcal{Y}$. The unknown input distribution is D on $\mathcal{X}$, and the labeled sample is

$$S = ((X_1, Y_1), \dots, (X_m, Y_m)), \quad X_i \stackrel{\text{i.i.d.}}{\sim} D, \quad Y_i = c(X_i).$$

A hypothesis class $\mathcal{H} \subseteq \{h : \mathcal{X} \rightarrow \mathcal{Y}\}$ is the set from which the learner may return a predictor. The distinction between a concept class $\mathcal{C}$ (possible targets) and a hypothesis class $\mathcal{H}$ (allowed outputs) matters:

- Realizable case: $c \in \mathcal{H}$;
- Agnostic case: $c \notin \mathcal{H}$.

For a hypothesis h , define the population and empirical 0–1 risks

$$R(h) = \mathbb{P}_{X \sim D}[h(X) \neq c(X)], \quad \hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{h(X_i) \neq c(X_i)\}.$$

Consistency of training and testing distributions: The training and testing data must be drawn from the same underlying distribution, i.e., $D_{\text{train}} = D_{\text{test}} = D$. The resulting generalization guarantee does not automatically apply under distribution shift, where $D_{\text{train}} \neq D_{\text{test}}$.

Definition 1.1 (PAC learning). *A class $\mathcal{C}$ is PAC-learnable by a learner A using hypotheses in $\mathcal{H}$ if there is a sample bound $m(\varepsilon, \delta)$ such that, for every target $c \in \mathcal{C}$, every distribution D , and every $\varepsilon, \delta \in (0, 1)$,*

$$m \geq m(\varepsilon, \delta) \implies \mathbb{P}_{S \sim D^m}[R(A(S)) \leq \varepsilon] \geq 1 - \delta.$$

The parameter ε controls accuracy; δ controls confidence.

Empirical-risk minimization Empirical-risk minimization (ERM) chooses

$$\hat{h} \in \arg \min_{h \in \mathcal{H}} \hat{R}_S(h).$$

In the realizable setting, a consistent learner returns $\hat{R}_S(\hat{h}) = 0$ since $\hat{R}_S(\hat{h}) \leq \hat{R}_S(c) = 0$. We treat ERM as an idealized oracle and focus on its statistical guarantees. In practice, some ERM problems are efficiently solvable, while others are computationally difficult or even NP-hard.

2 Finite classes: concentration and union bounds

2.1 The realizable consistent bound

Theorem 2.1 (Finite-class PAC bound). *Assume $\mathcal{H}$ is finite, $c \in \mathcal{H}$, and the learner returns a consistent hypothesis $\hat{h}$. Then, with probability at least $1 - \delta$,*

$$R(\hat{h}) \leq \frac{\log |\mathcal{H}| + \log(1/\delta)}{m}.$$

Consequently, $m \geq [\log |\mathcal{H}| + \log(1/\delta)]/\varepsilon$ implies $R(\hat{h}) \leq \varepsilon$ with probability at least $1 - \delta$.

Proof. Fix h with $R(h) > \varepsilon$. The events $\{h(X_i) \neq c(X_i)\}$ are independent and each has probability $R(h) > \varepsilon$. Hence

$$\mathbb{P}[\hat{R}_S(h) = 0] = (1 - R(h))^m \leq (1 - \varepsilon)^m \leq e^{-m\varepsilon},$$

where $1 - u \leq e^{-u}$ for $u \geq 0$. Let $\mathcal{H}_\varepsilon = \{h \in \mathcal{H} : R(h) > \varepsilon\}$. If a consistent learner has risk larger than ε , then some $h \in \mathcal{H}_\varepsilon$ has zero empirical error. Thus, by the union bound,

$$\mathbb{P}[R(\hat{h}) > \varepsilon] \leq \mathbb{P}\left[\bigcup_{h \in \mathcal{H}_\varepsilon} \{\hat{R}_S(h) = 0\}\right] \leq \sum_{h \in \mathcal{H}_\varepsilon} e^{-m\varepsilon} \leq |\mathcal{H}|e^{-m\varepsilon}.$$

Set the last quantity to at most δ and solve for m . □

Remark 2.2. *The $1/m$ rate is a realizable phenomenon. The dependence on confidence is only logarithmic, while halving ε approximately doubles the sample requirement. The term $\log |\mathcal{H}|$ is the information needed to distinguish among finitely many candidates.*

Example 2.3 (Discretized thresholds and conjunctions). *For $|\mathcal{H}| = 1001$, $\varepsilon = \delta = 0.05$, the sufficient sample size is $[\log(1001) + \log(20)]/0.05 \approx 198.1$, hence $m = 199$ suffices. For conjunctions over n Boolean variables, each variable can occur positively, negatively, or not at all, so $|\mathcal{H}| \leq 3^n$ and*

$$m \geq \frac{n \log 3 + \log(1/\delta)}{\varepsilon}.$$

2.2 Hoeffding and finite-class uniform convergence

Hoeffding's inequality for bounded variables in $[0, 1]$

Let $X_1, \dots, X_n$ be independent random variables with $X_i \in [0, 1]$. Define

$$\mu_i = \mathbb{E}[X_i], \quad Y_i = X_i - \mu_i, \quad \mu = \frac{1}{n} \sum_{i=1}^n \mu_i.$$

For every $t > 0$, we will prove

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq t\right) \leq 2e^{-2nt^2}.$$

Step 1: Bound the moment-generating function. For each i , define

$$M_i(\lambda) = \mathbb{E}[e^{\lambda Y_i}], \quad \psi_i(\lambda) = \log M_i(\lambda).$$

The exponentially tilted expectation is

$$\mathbb{E}_\lambda[f(Y_i)] := \frac{\mathbb{E}[f(Y_i)e^{\lambda Y_i}]}{\mathbb{E}[e^{\lambda Y_i}]}.$$

It reweights outcomes by $e^{\lambda Y_i}$. The corresponding variance is

$$\text{Var}_\lambda(Y_i) := \mathbb{E}_\lambda[Y_i^2] - (\mathbb{E}_\lambda[Y_i])^2.$$

Differentiating gives

$$\begin{aligned}\psi'_i(\lambda) &= \frac{M'_i(\lambda)}{M_i(\lambda)} = \mathbb{E}_\lambda[Y_i], \\ \psi''_i(\lambda) &= \frac{M''_i(\lambda)}{M_i(\lambda)} - \left(\frac{M'_i(\lambda)}{M_i(\lambda)}\right)^2 = \text{Var}_\lambda(Y_i).\end{aligned}$$

Tilting preserves the support $X_i \in [0, 1]$. Therefore,

$$\psi''_i(\lambda) = \text{Var}_\lambda(X_i) \leq \mathbb{E}_\lambda\left[\left(X_i - \frac{1}{2}\right)^2\right] \leq \frac{1}{4}.$$

Since $\psi_i(0) = \psi'_i(0) = 0$, Taylor's theorem implies

$$\psi_i(\lambda) = \frac{\lambda^2}{2}\psi''_i(\xi) \leq \frac{\lambda^2}{8},$$

where ξ lies between 0 and λ . Hence

$$\mathbb{E}[e^{\lambda Y_i}] \leq e^{\lambda^2/8}.$$

Step 2: Apply Markov's inequality and independence. For any $\lambda > 0$,

$$\begin{aligned}\mathbb{P}\left(\sum_{i=1}^n Y_i \geq nt\right) &\leq e^{-\lambda nt} \mathbb{E}\left[e^{\lambda \sum_{i=1}^n Y_i}\right] \\ &= e^{-\lambda nt} \prod_{i=1}^n \mathbb{E}[e^{\lambda Y_i}] \\ &\leq \exp\left(-\lambda nt + \frac{n\lambda^2}{8}\right).\end{aligned}$$

Choosing $\lambda = 4t$ yields

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \geq t\right) \leq e^{-2nt^2}.$$

Applying the same argument to $1 - X_i$ bounds the lower tail. A union bound completes the proof:

$$\boxed{\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq t\right) \leq 2e^{-2nt^2}.$$

For a fixed h , apply the lemma to $Z_i = \mathbf{1}\{h(X_i) \neq Y_i\}$. A union bound gives a simultaneous statement.

Proposition 2.4 (Finite-class uniform convergence). *With probability at least $1 - \delta$, every*

$h \in \mathcal{H}$ satisfies

$$|R(h) - \hat{R}_S(h)| \leq \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2m}}.$$

Proof. For a fixed h , choose $\gamma = \sqrt{\log(2|\mathcal{H}|/\delta)/(2m)}$ in Hoeffding, so the failure probability is $\delta/|\mathcal{H}|$. Summing over $|\mathcal{H}|$ hypotheses gives total failure probability at most δ . $\square$

Theorem 2.5 (Agnostic ERM bound for a finite class). *Let $\hat{h} \in \arg \min_{h \in \mathcal{H}} \hat{R}_S(h)$ and $h^* \in \arg \min_{h \in \mathcal{H}} R(h)$. On the uniform-convergence event,*

$$R(\hat{h}) \leq R(h^*) + 2\sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2m}}.$$

Proof. Let γ denote the deviation bound. Then

$$R(\hat{h}) \leq \hat{R}_S(\hat{h}) + \gamma \leq \hat{R}_S(h^*) + \gamma \leq R(h^*) + 2\gamma.$$

The first and third inequalities use uniform convergence; the middle one is ERM optimality. $\square$

3 Rademacher complexity

3.1 Definition

Let $\mathcal{G}$ be a family of functions $g : \mathcal{Z} \rightarrow \mathbb{R}$ and fix $S = (z_1, \dots, z_m)$. Let $\sigma_1, \dots, \sigma_m$ be independent uniform signs in $\{-1, +1\}$.

Definition 3.1 (Empirical Rademacher complexity).

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i g(z_i) \right].$$

The expected Rademacher complexity is

$$\mathfrak{R}_m(\mathcal{G}) = \mathbb{E}_{S \sim D^m} [\hat{\mathfrak{R}}_S(\mathcal{G})].$$

Writing $v_g(S) = (g(z_1), \dots, g(z_m)) \in \mathbb{R}^m$ gives

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{v \in A_S} \langle \sigma, v \rangle \right], \quad A_S = \{v_g(S) : g \in \mathcal{G}\}.$$

Thus complexity measures how well the class can align with random noise after seeing the sample.

Proposition 3.2 (Basic properties). *For a fixed S , if $\mathcal{G} \subseteq \mathcal{F}$, a is a fixed function, and $c \geq 0$, then*

$$\hat{\mathfrak{R}}_S(\mathcal{G}) \leq \hat{\mathfrak{R}}_S(\mathcal{F}), \quad \hat{\mathfrak{R}}_S(c\mathcal{G}) = c\hat{\mathfrak{R}}_S(\mathcal{G}), \quad \hat{\mathfrak{R}}_S(\mathcal{G} + a) = \hat{\mathfrak{R}}_S(\mathcal{G}),$$

and $\hat{\mathfrak{R}}_S(\mathcal{G}_1 + \mathcal{G}_2) \leq \hat{\mathfrak{R}}_S(\mathcal{G}_1) + \hat{\mathfrak{R}}_S(\mathcal{G}_2)$.

Proof. Monotonicity and positive homogeneity are immediate. For translation, the extra term $m^{-1} \sum_i \sigma_i a(z_i)$ has expectation zero. For subadditivity, use $\sup_{u,v} (A_u + B_v) \leq \sup_u A_u + \sup_v B_v$ inside the expectation. $\square$

Proposition 3.3 (Convexification does not increase complexity). *For finite convex hulls,*

$$\hat{\mathfrak{R}}_S(\text{conv}(\mathcal{G})) = \hat{\mathfrak{R}}_S(\mathcal{G}).$$

Proof. For a fixed sign vector, a convex combination $f = \sum_j \alpha_j g_j$ satisfies

$$\sum_i \sigma_i f(z_i) = \sum_j \alpha_j \sum_i \sigma_i g_j(z_i) \leq \sup_{g \in \mathcal{G}} \sum_i \sigma_i g(z_i).$$

The reverse inequality follows from $\mathcal{G} \subseteq \text{conv}(\mathcal{G})$; average over σ and divide by m . $\square$

3.2 The Rademacher generalization theorem

Theorem 3.4 (Expected-complexity bound). *Let $\mathcal{G}$ be a class of functions $g : \mathcal{Z} \rightarrow [0, 1]$. With probability at least $1 - \delta$ over $S \sim D^m$, simultaneously for all $g \in \mathcal{G}$,*

$$\mathbb{E}[g(Z)] \leq \frac{1}{m} \sum_{i=1}^m g(z_i) + 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

Proof. Define the worst one-sided gap

$$\Phi(S) = \sup_{g \in \mathcal{G}} \left(\mathbb{E}[g(Z)] - \widehat{L}_S(g) \right), \quad \widehat{L}_S(g) = m^{-1} \sum_i g(z_i).$$

Step 1 (bounded differences). If S, S' differ in one coordinate, then $|\Phi(S) - \Phi(S')| \leq 1/m$, because every g takes values in $[0, 1]$. McDiarmid's inequality gives, with probability at least $1 - \delta$,

$$\Phi(S) \leq \mathbb{E}_S \Phi(S) + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

Step 2 (ghost sample). Let $S' \sim D^m$ be independent. Since $\mathbb{E}[g] = \mathbb{E}_{S'} \widehat{L}_{S'}(g)$,

$$\mathbb{E}_S \Phi(S) = \mathbb{E}_S \sup_g \mathbb{E}_{S'} [\widehat{L}_{S'}(g) - \widehat{L}_S(g)] \leq \mathbb{E}_{S, S'} \sup_g [\widehat{L}_{S'}(g) - \widehat{L}_S(g)].$$

The inequality is $\sup_g \mathbb{E}_{S'} U_g \leq \mathbb{E}_{S'} \sup_g U_g$. *Step 3 (symmetrization).* Each pair (z_i, z'_i) is exchangeable. Randomly swapping the two elements with an independent sign σ_i preserves the joint distribution, hence

$$\begin{aligned} \mathbb{E}_{S, S'} \sup_g [\widehat{L}_{S'}(g) - \widehat{L}_S(g)] &= \mathbb{E}_{S, S', \sigma} \sup_g \frac{1}{m} \sum_i \sigma_i (g(z'_i) - g(z_i)) \\ &\leq \mathbb{E}_{S', \sigma} \sup_g \frac{1}{m} \sum_i \sigma_i g(z'_i) + \mathbb{E}_{S, \sigma} \sup_g \frac{1}{m} \sum_i (-\sigma_i) g(z_i) \\ &= 2\mathfrak{R}_m(\mathcal{G}). \end{aligned}$$

Combining the three steps yields the claimed inequality. $\square$

Proposition 3.5 (Empirical-complexity version). *With probability at least $1 - \delta$, all $g \in \mathcal{G}$ satisfy*

$$\mathbb{E}[g(Z)] \leq \widehat{L}_S(g) + 2\widehat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}.$$

Proof. Changing one sample point changes $\widehat{\mathfrak{R}}_S(\mathcal{G})$ by at most $1/m$: for every sign vector, changing one coordinate changes each correlation by at most $1/m$, and the same holds after taking a supremum and an expectation. McDiarmid therefore gives

$$\mathfrak{R}_m(\mathcal{G}) \leq \widehat{\mathfrak{R}}_S(\mathcal{G}) + \sqrt{\frac{\log(2/\delta)}{2m}}$$

with probability at least $1 - \delta/2$. Apply the expected-complexity theorem with confidence $\delta/2$ and use a union bound. $\square$

3.3 Classification as a loss-class problem

For $\mathcal{H} \subseteq \{h : \mathcal{X} \rightarrow \{-1, +1\}\}$ define the zero-one loss class

$$\ell \circ \mathcal{H} = \{(x, y) \mapsto \mathbf{1}\{h(x) \neq y\} : h \in \mathcal{H}\}.$$

Since $\mathbf{1}\{h(x) \neq y\} = (1 - yh(x))/2$ and signs are symmetric,

$$\mathfrak{R}_m(\ell \circ \mathcal{H}) = \frac{1}{2}\mathfrak{R}_m(\mathcal{H}), \quad \widehat{\mathfrak{R}}_S(\ell \circ \mathcal{H}) = \frac{1}{2}\widehat{\mathfrak{R}}_S(\mathcal{H}).$$

Thus, with probability at least $1 - \delta$, simultaneously for all $h \in \mathcal{H}$,

$$R(h) \leq \widehat{R}_S(h) + \mathfrak{R}_m(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2m}},$$

and an observable version replaces $\mathfrak{R}_m(\mathcal{H})$ by $\widehat{\mathfrak{R}}_S(\mathcal{H})$ and uses the corresponding empirical confidence constant.

4 Growth functions and Massart's lemma

For a binary class $\mathcal{H} \subseteq \{0, 1\}^{\mathcal{X}}$ and a finite input sample $S_X = (x_1, \dots, x_m)$, define its restriction

$$\mathcal{H}|_{S_X} = \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^m.$$

Although $\mathcal{H}$ may be infinite, $|\mathcal{H}|_{S_X}| \leq 2^m$. The growth function is

$$\Pi_{\mathcal{H}}(m) = \max_{x_1, \dots, x_m \in \mathcal{X}} |\mathcal{H}|_{S_X}|.$$

It counts distinct dichotomies, not parameter values.

Example 4.1 (Thresholds and intervals on $\mathbb{R}$). For $h_a(x) = \mathbf{1}\{x \geq a\}$ and ordered points $x_1 < \dots < x_m$, the positive block can start at any of $m + 1$ locations, so $\Pi(m) = m + 1$. For intervals $h_{a,b}(x) = \mathbf{1}\{a \leq x \leq b\}$, the positive block is empty or a nonempty contiguous block, giving

$$\Pi(m) = 1 + \binom{m+1}{2} = 1 + \frac{m(m+1)}{2}.$$

Lemma 4.2 (Massart's finite-set lemma). Let $A \subset \mathbb{R}^m$ be finite and let $r = \max_{a \in A} \|a\|_2$. Then

$$\frac{1}{m} \mathbb{E}_{\sigma} \sup_{a \in A} \langle \sigma, a \rangle \leq \frac{r \sqrt{2 \log |A|}}{m}.$$

Proof. For $\lambda > 0$, Jensen's inequality and $e^{\lambda \max_j u_j} \leq \sum_j e^{\lambda u_j}$ give

$$e^{\lambda \mathbb{E} \sup_{a \in A} \langle \sigma, a \rangle} \leq \mathbb{E} e^{\lambda \sup_{a \in A} \langle \sigma, a \rangle} \leq \sum_{a \in A} \mathbb{E} e^{\lambda \langle \sigma, a \rangle}.$$

By independence and Hoeffding's lemma, for each a ,

$$\mathbb{E} e^{\lambda \langle \sigma, a \rangle} = \prod_{i=1}^m \mathbb{E} e^{\lambda \sigma_i a_i} \leq \exp \left(\frac{\lambda^2 \|a\|_2^2}{2} \right) \leq \exp \left(\frac{\lambda^2 r^2}{2} \right).$$

Therefore

$$\mathbb{E} \sup_{a \in A} \langle \sigma, a \rangle \leq \frac{\log |A|}{\lambda} + \frac{\lambda r^2}{2}.$$

For $r > 0$ and $|A| > 1$, minimize the right side at $\lambda = \sqrt{2 \log |A|}/r$. The degenerate cases are immediate; divide by m . $\square$

Proposition 4.3 (Growth controls Rademacher complexity). *For $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$ and every sample S_X ,*

$$\widehat{\mathfrak{R}}_S(\mathcal{H}) \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}}, \quad \mathfrak{R}_m(\mathcal{H}) \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}}.$$

Proof. The evaluation vectors form a finite set A_S with $|A_S| \leq \Pi_{\mathcal{H}}(m)$ and each vector has Euclidean norm $\sqrt{m}$. Apply Massart's lemma with $r = \sqrt{m}$ and average over S . $\square$

5 VC dimension

5.1 Shattering

Definition 5.1 (Shattering and VC dimension). *A finite set $S_X = \{x_1, \dots, x_m\}$ is shattered by $\mathcal{H}$ if $\mathcal{H}|_{S_X} = \{0, 1\}^m$. The VC dimension is*

$$\text{VCdim}(\mathcal{H}) = \sup\{m : \Pi_{\mathcal{H}}(m) = 2^m\}.$$

If arbitrarily large finite sets are shattered, the VC dimension is infinite.

To prove $\text{VCdim}(\mathcal{H}) = d$, give (i) a constructive lower bound: one set of d points that is shattered; and (ii) a universal upper bound: no set of $d + 1$ points is shattered.

- One-sided thresholds on $\mathbb{R}$ have VC dimension 1;
- Intervals on $\mathbb{R}$ have VC dimension 2;
- Axis-aligned rectangles in $\mathbb{R}^2$ have VC dimension 4;
- Affine halfspaces in $\mathbb{R}^d$ have VC dimension $d + 1$.

Lemma 5.2 (Sauer's Lemma). *If $\text{VCdim}(\mathcal{H}) = d$, then for every integer $m \geq 0$,*

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i}.$$

Proof. We prove the statement by induction on m . The cases $d = 0$ or $m = 0$ are immediate. Fix an m -point set on which the maximum is attained, write $G = \mathcal{H}|_{S_X}$, and remove the last point to obtain S'_X . Let

$$G_1 = \{u \in \{0, 1\}^{m-1} : u \text{ occurs as a prefix in } G\},$$

$$G_2 = \{u \in \{0, 1\}^{m-1} : u0 \in G \text{ and } u1 \in G\}.$$

Every pattern in G contributes once to G_1 , and patterns whose prefix admits both last labels contribute one extra copy; hence $|G| = |G_1| + |G_2|$. The class of prefixes has VC dimension at most d , while G_2 has VC dimension at most $d - 1$: if G_2 shattered d prefix coordinates, then together with the last point those $d + 1$ coordinates would be shattered by G , contradicting VC dimension d . Therefore,

$$\Pi_{\mathcal{H}}(m) \leq \Pi_d(m - 1) + \Pi_{d-1}(m - 1).$$

Apply the induction hypothesis and Pascal's identity:

$$\begin{aligned} \Pi_{\mathcal{H}}(m) &\leq \sum_{i=0}^d \binom{m-1}{i} + \sum_{i=0}^{d-1} \binom{m-1}{i} \\ &= \sum_{i=0}^d \left[\binom{m-1}{i} + \binom{m-1}{i-1} \right] = \sum_{i=0}^d \binom{m}{i}. \end{aligned}$$

$\square$

Corollary 5.3 (Polynomial growth). *For $m \geq d \geq 1$,*

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i} \leq \left(\frac{em}{d}\right)^d.$$

Thus $\log \Pi_{\mathcal{H}}(m) \leq d \log(em/d)$.

Proof. The first inequality is Sauer–Shelah. The standard binomial estimate follows by bounding the partial sum by $(em/d)^d$ (or by the largest-term estimate together with $\binom{m}{d} \leq (em/d)^d$). $\square$

5.2 VC generalization bounds and sample complexity

Theorem 5.4 (VC bound via Rademacher complexity). *Let $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$ have VC dimension d and let $m \geq d \geq 1$. With probability at least $1 - \delta$, every $h \in \mathcal{H}$ satisfies*

$$R(h) \leq \hat{R}_S(h) + \sqrt{\frac{2d \log(em/d)}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

Proof. Start with the binary Rademacher generalization bound. The evaluation vectors have norm $\sqrt{m}$, and Sauer’s lemma gives $\Pi_{\mathcal{H}}(m) \leq (em/d)^d$. Massart’s lemma therefore yields

$$\mathfrak{R}_m(\mathcal{H}) \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}} \leq \sqrt{\frac{2d \log(em/d)}{m}}.$$

Substitute this estimate into the classification corollary. $\square$

Corollary 5.5 (ERM excess risk). *Let $\hat{h}$ minimize $\hat{R}_S$ and let h^* minimize R over $\mathcal{H}$. If*

$$\Gamma_S = \sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)|,$$

then

$$R(\hat{h}) - R(h^*) \leq 2\Gamma_S.$$

Proof.

$$R(\hat{h}) \leq \hat{R}_S(\hat{h}) + \Gamma_S \leq \hat{R}_S(h^*) + \Gamma_S \leq R(h^*) + 2\Gamma_S.$$

$\square$

Ignoring logarithmic factors, the estimation term scales as $\sqrt{(d + \log(1/\delta))/m}$. Solving the uniform-convergence requirement for a target deviation ε gives the agnostic sample-complexity scale

$$m = O\left(\frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon^2}\right).$$

In the realizable case, sharper consistent bounds have linear rather than quadratic dependence on $1/\varepsilon$.